NEW RESEARCH SESSION

AutoML Research Lab

A team of AI agents that designs, runs, critiques and improves machine learning experiments.
Choose a dataset and an objective, then watch the agents work.

1. Dataset

Drop a CSV file here
or choose a file from your computer

Upload CSV

Or use a bundled sample dataset

churn_sample.csv
Sample dataset · tabular, binary target

2. Target column

Task type

Classification

Target column

churned

The Data Scientist lead and its subagents profile the data and flag identifier or leakage-prone columns after you start.

3. Objective

Optimization metric

F1, positive class

Target score

0.70

Experiment budget

10

The system may stop at the budget without reaching the target score. That is a valid result.

Fixed protocol

- Train, validation and test sets are split once, with a fixed seed.
- Preprocessing is fitted on training data only.
- The sealed holdout is never shown to agents. It is opened once, at the end.
- Every run has retry, timeout, compute and experiment limits.

Start research session

RESEARCH CONTROL CENTER

AutoML Research Lab

Session s-0142 · illustrative sample data

Start | Pause at next checkpoint | Resume | Reset

Dataset	Target variable	Optimization metric	Target score	Experiment budget	Status
churn_sample.csv	churned	F1, positive class	0.70	10 experiments	Running

Live agent graph

Four leads and their subagents · driven by LangGraph events

Meta-supervisor · deterministic router, no LLM
experiment 5 of 10 · retries 0 of 3

Data Scientist — Completed
DataProfile · 18% positive class, 1 identifier flagged
✓ Data profiler · ✓ Leakage auditor · ✓ Feature analyst

Hypothesis Generator — Completed
ExperimentSpec for experiment 5
✓ Memory retriever · ✓ Hypothesis drafter · ✓ Novelty checker

ML Engineer — Running
Validating spec against parameter schema
● Config validator · Experiment runner · Error repair

Judge — Pending
Verdict on experiment 5
Metric analyst · Overfitting checker · Independent reviewer

Judge → Hypothesis Generator while budget remains and the target is not met

Experiment timeline

4 completed · 1 in progress

Each experiment links to the Judge verdict that motivated it

1 Logistic regression sets a baseline

logistic_regression · 1.2 s

Motivated by: Data Scientist profile, 18% positive class

C=1.0

Val F1	Precision	Recall	Train F1
0.41	0.58	0.32	0.44

Judge. Observation: recall 0.32 against precision 0.58 on the 18% positive class. Hypothesis: the model under-predicts the minority class.

Next action: Explore: class weighting

2 Class weighting may raise minority recall

logistic_regression · 1.4 s

Motivated by: Experiment 1 verdict, recall 0.32 is low

C=1.0 · class_weight=balanced

Val F1	Precision	Recall	Train F1
0.52	0.41	0.71	0.55

Judge. Observation: recall rose from 0.32 to 0.71, precision fell from 0.58 to 0.41. Hypothesis: a nonlinear model can recover precision.

Next action: Explore: tree ensemble

3 A random forest may recover precision

random_forest · 6.8 s

Motivated by: Experiment 2 verdict, precision fell to 0.41

n_estimators=300 · max_depth=12 · class_weight=balanced_subsample

Val F1	Precision	Recall	Train F1
0.58	0.55	0.61	0.97

Judge. Observation: train F1 0.97 against val F1 0.58. Hypothesis: depth 12 memorizes training rows; a regularized booster may generalize better.

Next action: Explore: gradient boosting

4 Boosting may generalize better than deep trees — Best so far

hist_gradient_boosting · 3.1 s

Motivated by: Experiment 3 verdict, overfitting at depth 12

learning_rate=0.1 · max_iter=200 · class_weight=balanced

Val F1	Precision	Recall	Train F1
0.63	0.60	0.66	0.71

Judge. Observation: val F1 0.63 and train F1 0.71, a smaller gap than experiment 3. Hypothesis: a lower learning rate with L2 may close the remaining 0.07 gap to the target.

Next action: Exploit: refine learning rate

5 Lower learning rate with L2 may narrow the gap — Validating

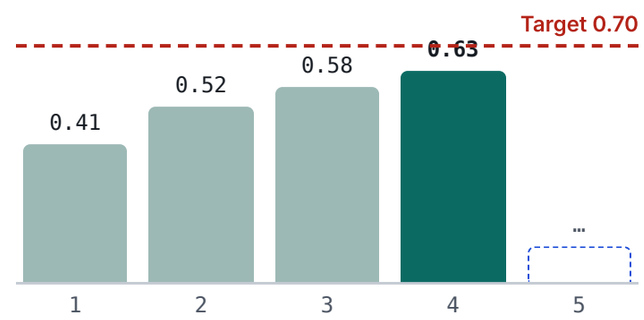
Motivated by: Experiment 4 verdict, gap to target 0.07

learning_rate=0.05 · max_iter=300 · l2_regularization=1.0 · class_weight=balanced

Metrics and the Judge verdict appear when the run finishes.

Validation F1 by experiment

Positive class · dashed line is the 0.70 target



Best versus previous model

	Exp 4 (best)	Exp 3
Val F1	0.63	0.58
Precision	0.60	0.55
Recall	0.66	0.61

Best model, precision and recall

Precision 0.60
Recall 0.66

Budget, duration and cost

5 experiments remaining of 10

Training time	Invalid configs	Run cost
12.5 s	0	$0.12

Agent activity log

Streaming over Server-Sent Events

```
10:42:03  DATA SCIENTIST     Detected 18% positive class rate
10:42:04  DATA SCIENTIST     Flagged customer_id as a likely identifier and excluded it
10:44:51  JUDGE              Experiment 3 train F1 0.97 against val F1 0.58: overfitting, recommend boosting
10:44:52  META-SUPERVISOR    Starting experiment 4 of 10
10:44:58  HYPOTHESIS         Proposed HistGradientBoosting, learning_rate 0.1, balanced class weight
10:45:00  ML ENGINEER        Configuration validated
10:45:03  ML ENGINEER        Model fitting completed in 3.1 s
10:45:05  JUDGE              Val F1 0.63, recall 0.66, precision 0.60; best so far, 0.07 below target
10:45:06  META-SUPERVISOR    Starting experiment 5 of 10
10:45:12  HYPOTHESIS         Proposed lower learning rate with L2 regularization
```

AutoML Research Lab: architecture overview

Control flow · Data flow · Event stream · Planned extension, not yet built

1 · Frontend: React, TypeScript, React Flow, Recharts

- **Setup and controls** — data, task, budget
- **Agent graph** — live node states
- **Experiment timeline** — hypotheses and verdicts
- **Metrics** — validation scores, budget
- **Activity log** — streamed events

commands

to all panels

2 · API and event streaming

- **FastAPI commands** — start, pause, resume, reset
- **Run worker** — background job per run
- **Model provider** — LangChain, set by env vars
- **SSE event stream** — agent, tool, metric events

start run

LLM calls

events

3 · LangGraph orchestration: four lead agents, three subagents each (details on the next screen)

Meta-supervisor, a deterministic LangGraph router · routes feedback, enforces budgets and stop rules, no LLM

accept or stop

- **Data Scientist** — profiles data, flags risks
  profiler · leakage auditor · feature analyst
- **Hypothesis Generator** — proposes testable experiments
  retriever · drafter · novelty checker
- **ML Engineer** — validates, runs, recovers
  validator · runner · error repair
- **Judge** — weighs evidence, decides
  metrics · overfitting · reviewer

new hypothesis or revise

state saved

reads memory

tool calls

writes results

writes verdicts

4 · State and memory: validation-level data only

- **Checkpointer** — graph state, resume
- **Working memory** — per-agent scratch
- **Findings board** — validated observations
- **Experiment history** — configs, metrics, critiques
- **Long-term lessons** — reusable insights

5 · Deterministic execution harness: no model-written code runs here

- **Execution backends** — pluggable runners, e.g. scikit-learn, PyTorch
- **Bounded workers** — concurrency cap, trial timeout (planned extension)
- **Validated tool layer** — inspect_data, validate_experiment, run_experiment, compare_experiments, get_history, save_artifact, generate_report
- **Policy and evaluation guard** — split, seeds, leakage checks, sealed holdout, budgets, retry limits

writes artifacts

logs runs

reads data

guards access

6 · Data and tracking

- **Artifact store** — models, plots, run files
- **MLflow tracking** — one run per experiment
- **Dataset store** — uploaded and bundled datasets
- **Sealed holdout** — opened once, by the harness

feeds report

best run

evaluated once

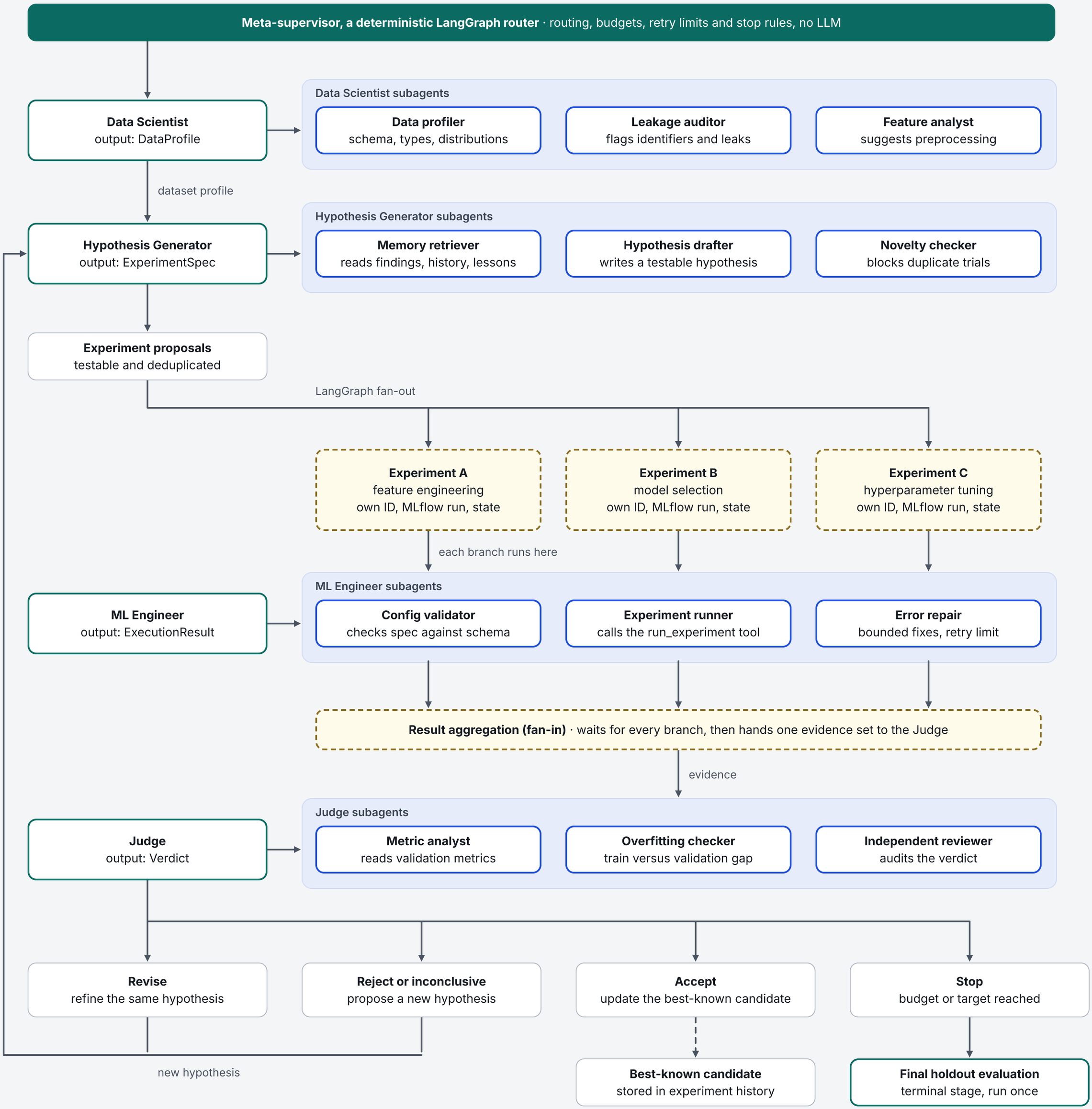
7 · Research output

- **Reproducible report** — config, seeds, metrics
- **Selected candidate** — best by validation
- **Final holdout evaluation** — terminal stage, run once

report

final evaluation

AutoML Research Lab: agent orchestration



AutoML Research Lab: bounded recovery and run state

Planned extension, not yet built

Bounded recovery loop for one experiment: the retry limit and the timeout come from the policy guard

ML Engineer
receives ExperimentSpec

Config validator
schema and ranges

Experiment runner
enforces trial timeout

Run succeeded?

yes

Judge
metrics and artifacts

no: failure or timeout

Error repair
one bounded fix proposed

Retries left?

yes: revalidate

no: retries exhausted

Trial record in graph state
experiment ID, attempt number, retry count, timeout status, failure reason

Terminal failure
status failed, reason kept

Failure report
reason, retries, logs

Meta-supervisor
stops the run or skips the branch

state saved after each step

LangGraph persistent state and checkpointer

Graph state
run, experiment and retry fields

Checkpointer
saved after every node

Run resume
continues from the last checkpoint

Process-failure recovery
restart picks up from a checkpoint

Human interruption
pause and edit at safe checkpoints (planned)

Research policy and evaluation guard: applies to every branch and every retry

Fixed protocol
train, validation and test split, fixed seeds

Leakage prevention
preprocessing fitted on training data only

Sealed holdout
never shown to agents, opened once at the end

Budgets and stop rules
max experiments, compute, wall-clock, max retries